Theo Gibbs

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Professional Appointments

2025 - | **Schmidt Science Fellow and Postdoctoral Researcher, New York University**
Adviser: Joy Bergelson

2021-2025 | **NSF Graduate Research Fellow in Ecology, Princeton University**

2018-2019 | **Research Assistant, University of Illinois at Urbana-Champaign**
Adviser: James O'Dwyer

Education

2019-2025 | **PhD in Quantitative and Computational Biology, Princeton University**
Advisers: Jonathan Levine and Simon Levin
Thesis: *Higher-order interactions and species coexistence in diverse ecological communities*

2019-2021 | **MA in Quantitative and Computational Biology, Princeton University**
Advisers: Jonathan Levine and Simon Levin

2014-2018 | **BS in Mathematics with Honors, University of Chicago**
Adviser: Stefano Allesina

Research Interests

My research combines theory and experiment to understand how diverse ecological communities stably coexist. I am especially interested in interactions that uniquely emerge in diverse communities, often called higher-order interactions. I conduct experiments with annual plants in the field and microbial strains in the lab. In parallel, I build and analyze theoretical models for how species interactions impact diverse coexistence.

Publications

Preprints and Manuscripts in Review

- **Gibbs, T.***, Mazzarisi, O.*, Fant, L., An, R., Grilli, J., Barabás, G. † & Song, C.† Label invariance: a guiding principle for ecological models. *bioRxiv* and in review at American Naturalist. (* and † denote equal contribution).

Published

12. **Gibbs, T.**, Dahlin, K., Brennan, J., Silveira, C.*, McManus, L.* Coexistence of bacteria with a competition-colonization tradeoff on a dynamic coral host. *Theoretical Ecology* (2026). (* denotes equal contribution).

11. Adler, P., Detto, M., Ellner, S., **Gibbs, T.**, Gold, Z., Leonard, S., Levine, J., Schiffer, A., Song, C., Stemkovski, M., Vahsen, M. & Levine, J. What have we learned from empirical applications of modern coexistence theory? *Ecology Letters* (2026).

10. Märkle, H., **Gibbs, T.** & Bergelson, J. Evolutionary implications of plant host interactions with a generalist pathogen. *Proceedings of the Royal Society B* (2026).

9. **Gibbs, T.***, Gold, Z.*, Oyler, H., Levine, J.,† Kraft, N.† Spatial clustering modifies competition in a diverse annual plant community. *Proceedings of the National Academy of Sciences* (2026). (* and † denote equal contribution).

8. **Gibbs, T.**, Levine, J. & Turcotte, M. Competitor-induced plasticity modifies the interactions and predicted competitive outcomes between annual plants. *Ecology* (2025).

7. **Gibbs, T.**, Gellner, G., Levin, S., McCann, K., Hastings, A. & Levine, J. When can higher-order interactions produce stable coexistence? *Ecology Letters* (2024).

6. **Gibbs, T.**, Levin, S. & Levine, J. Coexistence in diverse communities with higher-order interactions. *Proceedings of the National Academy of Sciences* (2022). **Honorable mention for the Outstanding Ecological Theory Paper Award from the Ecological Society of America.**

5. **Gibbs, T.**, Zhang, Y., Miller, Z. & O'Dwyer, J. Stability criteria for the consumption and exchange of essential resources. *PLoS Computational Biology* (2022).

4. Yamamichi, M., **Gibbs, T.**, & Levine, J. Integrating eco-evolutionary dynamics and modern coexistence theory. *Ecology Letters* (2022).

3. Levine, J., Levine, J., **Gibbs, T.** & Pacala, S. Competition for water and species coexistence in phenologically structured annual plant communities. *Ecology Letters* (2022).

2. D'Andrea, R.*, **Gibbs, T.*** & O'Dwyer, J. Emergent neutrality in consumer-resource dynamics. *PLoS Computational Biology* (2020). (* denotes equal contribution).

1. **Gibbs, T.**, Grilli, J., Rogers, T., & Allesina, S. Effect of population abundances on the stability of large random ecosystems. *Physical Review E* (2018).

Grants, Honors and Awards

2025 | Princeton University Ecology and Evolutionary Biology Service Award to the Levine Lab for the Plant Ecology Field Workshop

2025 | Banff International Research Station Pacific Institute of Mathematical Sciences Simons Travel Award – $986

2025 | Schmidt Science Fellow for Interdisciplinary Postdoctoral Research – $220,000

2025 | NSF Postdoctoral Fellowship: PRFB – $270,000 (declined)

2025 | Life Sciences Research Foundation (LSRF) Fellowship Finalist – $231,000 (declined)

2023 | Cassidy Yang Memorial Prize for excellence in academics and leadership in outreach from the Lewis-Sigler Institute at Princeton University – $1,500

2023 | Honorable Mention for the 2023 ESA Outstanding Ecological Theory Paper Award

2023 | Mary and Randall Hack '69 Graduate Award for Water and the Environment from Princeton University – $8,000

2021 | NSF Graduate Research Fellow in Ecology – $138,000

2019 | Institute Scholars Award in Quantitative and Computational Biology for first-year graduate students at Princeton University – $8,000

2018 | General Honors and Honors in Mathematics at the University of Chicago

Workshops, Talks and Seminars

Invited Workshops

2025 – Istituto Veneto di Scienze, Lettere ed Arti, Alma Dal Co School on Collective Behavior, Venice, Italy.
2024 – Hawaii Institute of Marine Biology, Theory of Microbial Symbiosis Workshop, Kaneohe, USA.

Invited Talks

2026 – BeyondTheEdge: Higher-Order Networks and Dynamics Seminar
2026 – International Initiative for Theoretical Ecology Seminar
2026 – University of Florida, Holt Lab Seminar
2026 – University of Florida, Biomathematics Seminar
2025 – Columbia University, Department of Ecology, Evolution and Environmental Biology Seminar
2025 – Virginia Tech, Mathematical Biology Seminar
2024 – Sun Yat-Sen University, Group of Professor Yuanzhi Li

2024 – New York University, Group of Professor Mingzhen Lu
2024 – Princeton University, Graduate Student Seminar for the Program in Applied and Computational Mathematics
2023 – The College of New Jersey, Colloquium in the Department of Mathematics and Statistics
2023 – RIKEN Interdisciplinary Theoretical and Mathematical Sciences Program, Biology Seminar
2023 – International Congress on Industrial and Applied Mathematics, "Hypernetworks and their dynamics in theory and applications" Minisymposium
2022 – SUNY Stony Brook, Group of Professor Rafael D'Andrea

Contributed Talks

2025 – Ecological Society of America, Modeling: Populations section
2024 – Princeton University, Theoretical Ecology Lab Tea
2024 – Ecological Society of America, Competition section
2023 – Ecological Society of America, Modeling: Communities, Disturbance, Succession section
2023 – Princeton University, Theoretical Ecology Lab Tea
2023 – Princeton University, QCB Graduate Student Colloquium
2023 – American Physical Society March Meeting, Ecological Dynamics session
2022 – British Ecological Society, Theoretical and Computational Ecology section
2022 – Ecological Society of America and Canadian Society for Ecology and Evolution Joint Meeting, Modeling section
2022 – Princeton University, Theoretical Ecology Lab Tea
2022 – American Physical Society March Meeting, Ecological and Evolutionary Dynamics session
2022 – Princeton University, QCB Graduate Student Colloquium
2021 – Princeton University, Theoretical Ecology Lab Tea
2021 – Princeton University, QCB Graduate Student Colloquium
2020 – Princeton University, Theoretical Ecology Lab Tea
2019 – University of Tokyo, Towards an Integration of Diverse Concepts in Community Ecology Workshop
2018 – NetSci (Annual Conference of the Network Science Society), Ecology Session

Outreach and Service

2023-2025 | **Graduate Student Representative on the QCB Climate Committee, Princeton University**
Met regularly with administrators, postdocs and faculty to improve the climate in QCB

2022-2025 | **Instructor for the Plant Ecology Field Workshop, Princeton University**
Brought local community college and Princeton students on a field work trip to a marsh ecosystem

2021-2024 | **QCB Peer Mentor, Princeton University**
Co-organized and participated in a mentoring program for beginning graduate students

2020-2025 | **Access, Diversity and Inclusion Outreach Student Ambassador, Princeton University**
Recruited students from under-represented backgrounds with the ADI team at Princeton

2020-2025 | **QCB Virtual Open House, Princeton University**
Organized the first (and now annual) virtual QCB open house to recruit applicants from non-traditional and under-represented backgrounds

2023-2024 | **Ecology and Evolutionary Biology (EEB) Mentor, Princeton University**
Mentor for under-represented students who are interested in graduate school in EEB

2021 | **Theoretical Ecology Lab Tea Organizer, Princeton University**

2020-2021 | **Princeton Online Tutoring Network, Princeton University**
Provided free tutoring assistance to under-served K-12 students in the local community

Mentoring and Teaching Experience

Teaching Assisantships

2023: Mathematical Methods in Biology and Medicine with Professor Corina Tarnita.
2022: Theoretical Ecology with Professor Simon Levin.
2016-2017: Tutored university undergraduate students in any math class.
2015-2016: Calculus I-III-III.

Graduate Students

Jiayu Zhang (2023-2024) on sparse higher-order interactions. Zachary Gold (2024-2025) on experimental measurements of higher-order interactions.

Undergraduate Students

Noah Egan (2023-2024) and Krishna Girish (2022-2023) on spatial higher-order interactions. Yifan Zhang (2020-2022) on microbial theory which produced a co-authored paper.

Reviewer

PNAS, PloS Computational Biology, Communications Biology, American Naturalist, Physical Review E, Scientific Reports, Ecology Letters, Theoretical Ecology, Physical Review X, Nature Communications, Science.

Technical Skills

- Programming in R, Python, C, LaTeX & Git
- Bayesian statistical analysis
- Statistical physics theory applied to ecology
- Dynamical systems and stochastic processes
- Field experiments with annual plants
- Analyzing microbial sequence data